

Exp No: 30

RTOS TASK SCHEDULING

```
WhatsApp
mithra@DESKTOP-6DUTPF3: task_scheduling.cpp
GNU nano 8.7.1
#include <systemc.h>
SC_MODULE(rtos_scheduler) {
void task1() {
while(true) {
cout << "Task1 Running at "
<< sc_time_stamp() << endl;
wait(1, SC_SEC);
}
}
void task2() {
while(true) {
cout << "Task2 Running at "
<< sc_time_stamp() << endl;
wait(1, SC_SEC);
}
}
SC_CTOR(rtos_scheduler) {
SC_THREAD(task1);
SC_THREAD(task2);
}
};
int sc_main(int argc, char* argv[]) {
rtos_scheduler obj("Scheduler");
sc_start(6, SC_SEC);
return 0;
}
[ Read 26 lines ]
G Help Write Out Where Is Cut Execute Location M-U Undo M-A Set Mark
X Exit Read File Replace AU Paste AJ Justify Go To Line M-E Redo M-6 Copy
```

```
WhatsApp
mithra@DESKTOP-6DUTPF3:
SystemC 3.0.2-Accellera --- Nov 5 2025 12:44:46
Copyright (c) 1996-2025 by all Contributors,
ALL RIGHTS RESERVED
Low Priority Interrupt at 1 s
High Priority Interrupt at 2 s
mithra@DESKTOP-6DUTPF3:~/exp29$ nano interrupt_handling.cpp
mithra@DESKTOP-6DUTPF3:~/exp29$ mkdir ~/exp30
mithra@DESKTOP-6DUTPF3:~/exp29$ cd ~/exp30
mithra@DESKTOP-6DUTPF3:~/exp30$ pwd
/home/mithra/exp30
mithra@DESKTOP-6DUTPF3:~/exp30$ nano task_scheduling.cpp
mithra@DESKTOP-6DUTPF3:~/exp30$ g++ task_scheduling.cpp -lsystemc -o task_scheduling
mithra@DESKTOP-6DUTPF3:~/exp30$ ./task_scheduling

SystemC 3.0.2-Accellera --- Nov 5 2025 12:44:46
Copyright (c) 1996-2025 by all Contributors,
ALL RIGHTS RESERVED
Task1 Running at 0 s
Task2 Running at 0 s
Task1 Running at 1 s
Task2 Running at 1 s
Task1 Running at 2 s
Task2 Running at 2 s
Task1 Running at 3 s
Task2 Running at 3 s
Task1 Running at 4 s
Task2 Running at 4 s
Task1 Running at 5 s
Task2 Running at 5 s
mithra@DESKTOP-6DUTPF3:~/exp30$
```